

Appendix - Bridge the Gaps: Heterogeneous Attributed Graph Clustering via Quaternion Representation Learning

I. SUPPLEMENTARY EXPERIMENTAL RESULTS

This section contains supplementary experiments providing additional validation of the proposed method's effectiveness.

A. Statistical Significance Analysis

To further verify the reliability of the observed improvements, the Wilcoxon signed-rank test is conducted on ACC, NMI, and ARI between AGREE and the compared methods. The corresponding p -values are shown in Fig. 1, where smaller p -values indicate stronger statistical evidence for performance differences. It can be observed that, on the mixed-type attributed datasets, AGREE shows statistically significant improvements over all compared methods at the 95% confidence level. On the attributed graph datasets, the superiority of AGREE is also significant against most baselines, while against several strong contrastive methods, i.e., CCGC and SCDGN, the differences remain significant mainly at the 90% confidence level. MAGI is also highly competitive due to its stronger discriminative objective. The competitiveness of AGREE in these comparisons further highlights the expressive power of quaternion representation learning in capturing feature interactions and integrating attribute semantics with graph structure. Overall, these results further confirm that the advantage of AGREE is statistically reliable rather than caused by random variation.

B. Efficiency Analysis

To evaluate the efficiency of AGREE under increasing graph size, node-scaling experiments are conducted on sampled subgraphs of ogbn-arxiv from the Open Graph Benchmark (OGB) [1]. The node ratio is gradually increased from 1% to 12%, corresponding to 1700, 8500, 13500, 16900, and 20300 nodes. Runtime, peak GPU memory, and peak CPU memory are compared with several competitive baselines (i.e., CCGC, EGAE, CONVERT, SCDGN, MAGI, and CDC). Fig. 2 reports (a) runtime and (b) peak memory under node scaling, respectively.

As the node scale increases, the runtime and memory consumption of AGREE both rise steadily. Although AGREE is slower than CDC, it remains competitive with CCGC and MAGI, and is more efficient than EGAE, CONVERT, and SCDGN over the evaluated scales. In terms of memory, SCDGN exhibits the highest GPU and CPU usage, while the memory usage of AGREE also increases with graph size due to the dense adjacency-related operations in graph construction and reconstruction. Nevertheless, AGREE remains below the

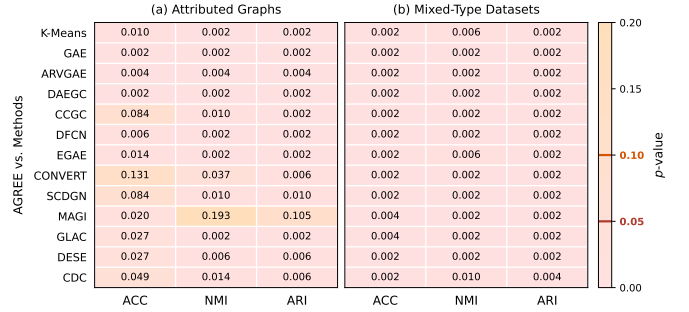


Fig. 1: Wilcoxon signed-rank test p -values between AGREE and the compared baselines on attributed graph datasets and mixed-type attributed datasets for ACC, NMI, and ARI.

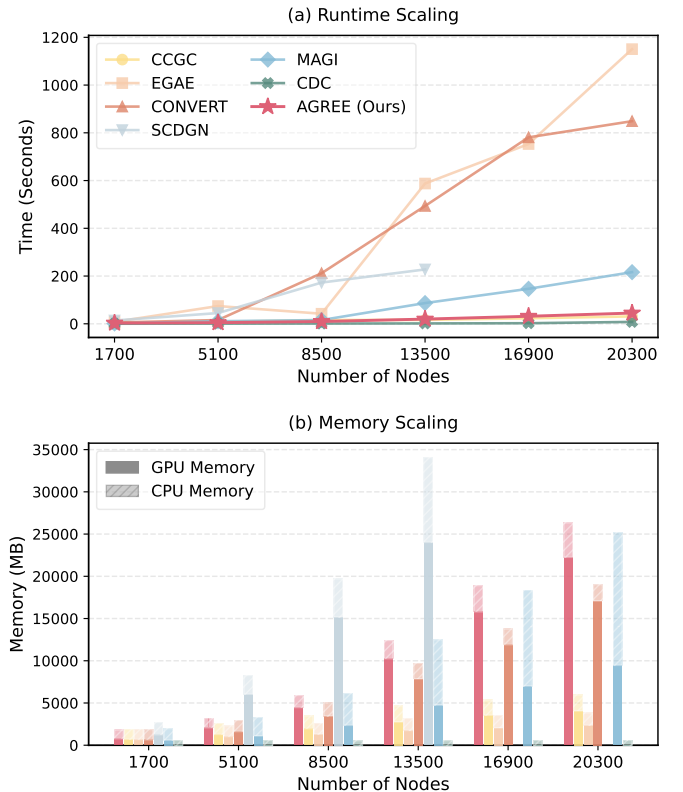


Fig. 2: Efficiency comparison of different methods with respect to the number of nodes. (a) Runtime using observed data points only. (b) Peak memory usage with GPU and CPU components shown separately. Missing entries indicate unavailable measurements, typically caused by OOM or runtime errors.

most memory-intensive baseline, which is consistent with the complexity analysis in Remark 6 of the main paper. These results suggest that sparse or blockwise graph operations could further improve the scalability of AGREE.

REFERENCES

- [1] W. Hu, M. Fey, M. Zitnik, Y. Dong, H. Ren, B. Liu, M. Catasta, and J. Leskovec, "Open Graph Benchmark: Datasets for machine learning on graphs," in *Advances in Neural Information Processing Systems*, 2020.